

HsiaoMan Liu

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Business Intelligence Engineer and Data Analyst

EDUCATION

University of Washington, Information School

Master of Science in Information Management

Specialization: Business Intelligence, Data Science, People Analytics and Program/Project Management | GPA: 3.8

Seattle, WA

June 2026

University of Washington, School of Public Health

Bachelor of Arts in Food Systems, Nutrition and Health

Minor: Data Science and Business Administration | GPA: 3.8 (Annual Dean's List 2022-2024)

Seattle, WA

June 2024

PROFESSIONAL EXPERIENCE

People Analytics Researcher

Sep 2025 – Present

University of Washington People Analytics Insights Lab

Seattle, WA

- Architected ELT pipelines for 1,000+ employee records using Python (Pandas, NumPy), improving HR data quality by 40% and enabling accurate predictive modeling of performance trends and attrition risk.
- Conducted time-series analysis on employee performance, tenure, and engagement metrics to identify attrition patterns, delivering actionable insights that informed talent retention strategies and workforce planning.
- Led end-to-end workforce analytics research in a Web 3 company, statistically testing variables (ANOVA, regression, SEM) to framework engagement and new employee performance, translating findings into executive presentations.

Operations Student Assistant

April 2023 – June 2025

University of Washington Department of Chemistry

Seattle, WA

- Improved data accessibility by 60% and enabled compliance reporting for HR and Finance Teams by spearheading digital transformation initiative, migrating 5000+ legacy HR records into structured database system.
- Reduced approval wait time by 50% (14 to 7 days) by querying SQL tables analyzing cycle time data and user surveys, identifying bottlenecks and implementing ServiceNow ticketing system to eliminate manual workflow delays.
- Optimized financial reconciliation processes by analyzing Workday ERP system data using advanced Excel (VLOOKUP, INDEX/MATCH, Power Query), automating invoice generation and reducing data discrepancies by 35%.

Data and Business Intelligence Analyst

July 2022 – April 2023

Fulllife Veterinary Hospital

Remote

- Built Tableau dashboards analyzing client scheduling and visit data to identify peak cancellation windows and idle time slots, informing schedule restructuring that reduced no-shows by 32% and improved on-time appointment rates by 25%.
- Performed requirements gathering, data modeling, and refinement using SQL and Snowflake from 10+ sources to achieve company results.
- Reduced time spent on data entry and information gathering from 4 weeks to 14 days by creating ETL pipelines, staging, dimension and fact tables to support dashboard building and data analytics.

PROJECT EXPERIENCE

Global Bike Inc. Business Intelligence Platform (2026) (maisieliu.github.io/work/gbi-business-intelligence)

- Improved forecast accuracy for discount strategies by developing predictive models on GBI transaction data analyzing sales, revenue, and distribution patterns across global markets using advanced Excel and Data Solver.
- Engineered data warehouse infrastructure in Snowflake, designing dimensional schemas and fact tables that consolidated sales, inventory, and distribution data for enterprise-scale analytics.
- Architected semantic models in Power BI using DAX and Power Query to transform raw transaction data into executive-ready business dashboards, enabling data-driven decision-making for sales and operations teams.

Box Office Forecasting: Predicting Movie Revenue (2025)

- Engineered predictive models using Python (scikit-learn) on IMDb dataset with over 1000 records, comparing kNN, Random Forest, Decision Trees, and Logistic Regression; used R-squared and MSE to validate results.
- Performed feature engineering (categorical encoding, standardization, interaction terms) and visualized model insights using Seaborn/Matplotlib, identifying budget, genre, and release season as top revenue drivers.

SKILLS

- **Programming & Database:** Python (Pandas, NumPy, scikit-learn), R, SQL, SAS, VBA | MySQL, PostgreSQL, Oracle SQL, Snowflake, Databricks, AWS (RDS, S3, Redshift), BigQuery
- **BI & Visualization:** Tableau, Power BI, Looker, Cognos BI, Domo, Qlik, Excel, Streamlit, Shiny, Alteryx
- **Analytics & Statistics:** ETL/ELT, Data Modeling, Statistical Testing (ANOVA, t-test, chi-square), Hypothesis Testing, A/B Testing, Regression, Predictive Modeling, Time-Series Analysis, Data Warehousing, Data Governance, Machine Learning
- **Project Management:** Agile/Scrum, JIRA, Asana, Trello, Stakeholder Management, Product Analytics